

Thinking Harder on Incidence

Bottom line. Reviewers find it strange that interchange-fee incidence in the paper falls almost entirely on different *consumer groups* when, statutorily, merchants pay the fees and banks collect the rewards. It's straightforward to extend our estimates to show aggregate consumer welfare, merchant profits, and network (which includes Visa/MC and the issuers) profits. The key inputs are then how much retailers pass-through interchange into prices (ρ^P) and how much issuers pass-through interchange into rewards (ρ^R). Sales-tax pass-through evidence puts $\rho^P \approx 1$, and retail-banking evidence puts ρ^R high but below one. Our current estimates thus likely understate the reduction in aggregate consumer welfare due to high interchange fees.

The formula. Recall that the generalized welfare formula is

$$\frac{1}{\lambda_k} \frac{dV_k}{dx_l} \approx \mu_k^{-1} \left(-\rho^P \mathbb{E}_R \left[\mu_{jk} \mu_{jl} \frac{\partial \tau_{jl}}{\partial x_l} \right] + \rho^R \mathbb{1}\{k = l\} \mathbb{E}_R \left[\mu_{jk} \frac{\partial \tau_{jk}}{\partial x_k} \right] \right).$$

The formula tracks consumer k 's welfare; the rest of each interchange dollar goes elsewhere. Of each dollar of merchant fee revenue, ρ^P is passed through to prices (the first term) and the remaining $(1 - \rho^P)$ stays with merchants as absorbed margin. Of each dollar of issuer interchange revenue, ρ^R flows out to cardholders as rewards (the second term) and the remaining $(1 - \rho^R)$ stays with banks as profit.

Summing the consumer formula across k (weighted by spending share μ_k) collapses the μ_{jk} weights via $\sum_k \mu_{jk} = 1$. Per dollar of total economy revenue, the three aggregate dollar flows from a change dx_l are

$$\begin{aligned} \text{merchants:} & \quad - (1 - \rho^P) \mathbb{E}_R \left[\mu_{jl} \frac{\partial \tau_{jl}}{\partial x_l} \right], \\ \text{banks:} & \quad + (1 - \rho^R) \mathbb{E}_R \left[\mu_{jl} \frac{\partial \tau_{jl}}{\partial x_l} \right], \\ \text{consumers (summed):} & \quad - (\rho^P - \rho^R) \mathbb{E}_R \left[\mu_{jl} \frac{\partial \tau_{jl}}{\partial x_l} \right], \end{aligned}$$

and the three rows sum to zero. If in reality $\rho^P \approx 1$ but $\rho^R < 1$, then these formulas suggest that we are undercounting the welfare gain of payment networks from high interchange fees and thus undercounting the aggregate loss borne by consumers.

The case for $\rho^P \approx 1$ (prices). *Sales taxes are the right analogue.* Interchange is an ad-valorem fee paid by merchants on every card transaction. The closest empirical literature is sales-tax pass-through. Butters et al. (2022) use national-firm scanner data and find that retailers respond to local cost shocks roughly one-for-one, the relevant pass-through margin for a sector-wide ad-valorem shock like interchange.

“Pricing in levels” still implies $\rho^P \approx 1$ for small shocks. The most influential recent finding of *incomplete* cost pass-through is Sangani (2026), who documents that retailers pass cost shocks into prices roughly one-for-one in dollar terms rather than in percent terms. For interchange differentials of a few tens of basis points, the levels and percent frames coincide arithmetically.

Avoidance attenuation does not apply here. The other large class of incomplete pass-through, exemplified by Harding et al. (2012) on cigarette taxes near state borders, operates through consumers shifting where they shop in response to the implicit tax. The avoidance margin is essentially absent for interchange. Virtually every brick-and-mortar merchant accepts cards and pays interchange, so consumers cannot duck the wedge by changing stores.

Direct evidence in our paper. The Durbin pass-through estimates (Section on Fact 4) deliver the same conclusion for interchange itself. A 1pp change in interchange moves retail prices roughly one-for-one within two years.

The case for $\rho^R \lesssim 1$ (rewards). *RBA evidence.* Chang et al. (2005) study the 2003 Reserve Bank of Australia interchange cap, a sharp exogenous reduction in issuer interchange revenue. They document that issuers offset roughly half the lost revenue through higher annual fees and roughly half through lower rewards generosity. The net consumer-incidence offset is close to full pass-through, consistent with ρ^R near one.

U.S. Durbin evidence is mixed but, on net, high. The published version of Mukharlyamov and Sarin (2025) reports pass-through to debit cardholders around 20%, while earlier working-paper versions found it closer to 100%. Kay et al. (2018) find that issuers offset roughly 90% of the lost interchange revenue through changes in deposit account terms. The estimates have not converged, but the modal point is high. We should report that honestly rather than pin down $\rho^R = 1$.

Accounting bound. Industry data put credit-card rewards at roughly 80% of total interchange revenue. With any positive operational cost of running a card program, the 80% ratio implies near break-even economics on the marginal interchange dollar, consistent with marginal ρ^R near one even when average issuer margins are positive.

Marginal versus average is the right distinction. The sufficient statistic depends on *marginal* pass-through, not average bank profitability. If Chase always retains a fixed 30 bps spread between

interchange and rewards, the constant wedge cancels in $dV_k/d\tau_c$. What matters is whether the next dollar of interchange revenue is paid out as rewards. Competitive issuer markets, the 80% rewards-to-interchange ratio, and the RBA and Kay et al. quasi-experimental evidence all point to a marginal pass-through near one.

Implication for the framing. With $\rho^P \approx 1$ and ρ^R modestly below one, the cash- and debit-user loss stays around $\mathbb{E}_R[\mu_{jk}\mu_{jc}]$ and is unchanged from the baseline. The credit-user gain shrinks in proportion to ρ^R , and banks pocket the rest. The headline transfer to credit users therefore *understates* the loss to cash and debit users relative to a more realistic world where banks part of interchange fees as higher profits.

Other Thoughts I Couldn't Make Tight

If consumer payment choice is endogenous, our current sufficient statistic actually excludes a large aggregate loss to consumers due to the fact that consumers overuse high reward payment methods. This is the force explored in my JMP. Intuitively, consumers collectively would prefer to use debit cards and pay low retail prices, but they individually face huge incentives to distort their payment choice towards credit cards because nobody internalizes the effect on retail prices. There is some scope to essentially write a sufficient statistic version of my JMP welfare results for consumers.

If consumer payment choice and merchant acceptance are endogenous, networks actually don't lose much profit on the margin from a small reduction in interchange fees. That is because networks are already at an envelope condition with respect to the interchange fee. The only first-order forces are again on consumers and merchants. I think we can exploit this in some broader sufficient statistic to reason about small, marginal changes in interchange. This could be combined with the sufficient statistic above to make even stronger statements about small reforms to interchange.

Fraud is covered by issuer banks if it's in-person, covered by merchants if it's online. But no matter the costs, no obvious reason to make the merchant pay for it. If the card is creating benefits for the consumer, we should have the consumer pay.

Even if consumers get benefits due to expanding credit access from interchange, there's not a clear reason why merchants need to pay for that. Consumers should be willing to pay.

References

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